

ADR-007 Companion — Dual-Path (“2-Prong”) Model: Variance & Impact Analysis

Companion analysis to ADR-007 (CRG/CR Deployment Model)

Field	Value
Document	Variance & Impact Analysis — Dual-Path CRG Model
Parent ADR	ADR-007 — CRG/CR Deployment Model
Status	ANALYSIS — for review before ADR-007 is moved from PROPOSED to ACCEPTED
Date	2026-09-09
Baseline	Requirements Baseline v2.4 (7 Sep 2026) — re-read for this analysis
Author(s)	ACRME Platform Architecture
Purpose	Answer: What does the 2-prong model change across the overall design? List each variance, plan to address each, and call out specifically what breaks.

Label conventions: [Baseline Requirement] = normative text in v2.4; [Verified Azure Fact – FC-*nn*] = confirmed against MS Learn (see ADR-007 §2); [Assumption] = carries risk; [Variance] = a delta this model introduces vs the current baseline/design corpus.

1. Framing Check — Is It One Model or Two? (Read This First)

Your phrasing is “2-prong model ... more modern focused.” Before listing variances, the framing must be pinned down, because the variance set is different depending on which of two interpretations we adopt. **Being critical of the hypothesis is part of the task, so this is called out explicitly rather than assumed away.**

Interpretation	What it means	ADR-007 position	Consequence for region selection
(A) Two parallel operating models	A distinct modern operating model and a distinct heritage operating model, each with its own structure, naming, life-cycle, and engine logic.	Rejected. ADR-007 §3.3 explicitly finds “ no separate operating model for modern vs heritage.”	Would add complexity — two structures the engine must reason about.
(B) One unified structural model, two access paths	A single CRG structure (CAP-023), naming (OPS-006/C-12), and reconciliation logic. Only the CRG-resolution access path differs: shared (modern + remediated) vs local (un-remediated heritage).	Recommended (Option C).	Does not touch region selection. The path branch lives in CRG resolution, downstream of region/zone selection.

This analysis adopts interpretation (B) — that is the “2-prong” your statement supports (“shouldn’t add complexity to region selection”). Every variance below is scoped to interpretation (B). If the enterprise instead wants interpretation (A), variances V-05, V-07, and V-14 expand materially and region selection would be impacted; that path is not recommended and is flagged at the end (§7, Open Decision OD-1).

Net answer to “is it more modern focused?” Yes — the shared prong is the strategic default for modern (environment-aligned) subscriptions; the local prong is a **coexistence/fallback path** that keeps heritage workloads running unchanged until remediated. In production **today**, the local prong is the only production-safe path until DEP-001 (Preview→GA) and POC-001 (quota) clear — the shared prong is aspirational until then (see V-10, V-11).

2. Region-Selection Guarantee (Your Specific Concern)

[Baseline Requirement] Region/zone selection is driven by the placement scoring pipeline over the configurable catalogue (REG-001), the Hard Constraints, ENV-003 separation, and PLC-006/PLC-010/PLC-010a co-location rules. **Nothing in region selection references CRG ownership or sharing.**

The engine has two distinct layers; the 2-prong branch is confined to Layer 2:

Layer	Question	Inputs	2-prong impact
L1 — Region/zone selection (ADR-001)	Which region + AZ?	REG-001 catalogue, geography model (US 3-region / others 2-region), ENV-003, PLC-010/010a	None — path-agnostic
L2 — CRG resolution (ADR-007 §10.3)	Which CRG, owned by which subscription?	(domain, region, track) → provider sub → CRG (shared) or local CRG in consumer sub (local)	This is the only place shared/local branches

In the §10.3 consumer-assignment algorithm, **region** is an input, not an output — L1 has already decided it before L2 runs. A heritage (local) and a modern (shared) workload in the same geography select the **identical region + AZ**; they differ only in who owns the CRG in that zone.

Variance V-06 (new, protective): ADR-007 relies on this ordering but never states it as a hard invariant. We add:

INV-R1 (proposed): Region/zone selection (ADR-001 / PLC-006) is path-agnostic. The shared-vs-local CRG determination occurs only in the CRG-resolution layer, which consumes region as an input. No modern/heritage classification, provider-subscription lookup, or sharing-grant state may be an input to region or zone selection.

This makes your requirement enforceable and prevents a future change from leaking heritage logic upward into region selection.

3. Complete Variance Register

Each variance is classified as **BREAKING** (contradicts or invalidates existing normative text / design artifacts), **ADDITIVE** (net-new; nothing existing becomes wrong), or **RECONCILE** (existing docs describe a narrower world and must be updated to stay consistent, but were not wrong at the time).

ID	Variance	Source of truth affected	Type	Gate
V-01	Sharing limit semantics: “~100 per tenant” → exactly 100 consumer subs per CRG + sharding	CAP-013	BREAKING (text + math)	—
V-02	Provider subscription owns CRGs, separate from workload subscription	CAP-023, deployment-lay-out, provisioning-plan, RBAC guide	BREAKING (ownership locus)	POC-006
V-03	Sharing-grant lifecycle (grant/revoke/audit) as a first-class engine duty	(none — gap)	ADDITIVE (new CAP-026)	DEP-001
V-04	MG-scoped sharing grants for NFR-004 scale	(none — gap)	ADDITIVE (CAP-013 clause)	POC-003
V-05	Engine must resolve two CRG paths (shared vs local) + CRG-type flag in a shard registry	Reconciliation design, UML class model	ADDITIVE (new state + branch)	—
V-06	Region-selection invariant INV-R1 (path-agnostic)	ADR-001 principles	ADDITIVE (protective)	—
V-07	Heritage/modern classification + heritage excluded from sharing until re-mediated	(none — new taxonomy)	ADDITIVE (A-01/ A-02 → firm rule)	OD-1
V-08				—

ID	Variance	Source of truth affected	Type	Gate
	ENV-003 enforced at a new surface : the sharing-grant boundary (grant audit), not only intra-subscription non-mixing	ENV-003 enforcement design	RECONCILE (mechanism extension)	
V-09	Position taken: dedicated DR provider subscription	POC-006 (open)	RECONCILE (pending POC)	POC-006
V-10	Shared-prong viability depends on quota behaviour (provider-vs consumer-scoped)	POC-001 (open) — cost model	DEPENDENCY (blocks shared prong)	POC-001
V-11	Shared prong is Public Preview , not production-safe until approved/GA	DEP-001 (open)	DEPENDENCY (blocks shared prong)	DEP-001
V-12	Naming convention extension for provider subs + shard counter	OPS-006 / C-12	RECONCILE (additive tokens)	—
V-13	Downstream design corpus describes single-path CRG ownership	6 design docs + 3 diagrams	RECONCILE	—

4. What BREAKS — Explicit Call-Out

Only three items genuinely break something already written or already assumed. Everything else is additive or a doc-reconciliation. These three must be resolved before ADR-007 moves to ACCEPTED.

B-1 — CAP-013 limit semantics (V-01)

- **Current baseline text:** “...up to ~100 subscriptions per tenant.”
- **What breaks:** The number is attached to the **wrong scope** (tenant) and marked approximate. [Verified Azure Fact – FC-03] the real limit is **exactly 100 consumer subscriptions per CRG** — a hard platform limit, not per-tenant and not a Preview soft cap. [FC-02] the sharing unit is the CRG, not the CR.
- **Specifically what it invalidates:** Any capacity-planning arithmetic, POC scenario, or sizing statement that treated “100” as a tenant-wide ceiling. Under the correct semantics, a single tenant can serve **hundreds** of consumer subscriptions (NFR-004) by adding CRGs/shards — but no single CRG may exceed 100 consumers. Docs that read “~100/tenant” imply the platform cannot meet NFR-004, which is false and would mis-scope the sharding requirement.
- **Fix:** Adopt the CAP-013 correction already drafted in ADR-007 §15.1 (per-CRG limit + mandatory workload-domain sharding + CRG-is-the-unit). See remediation task T-01.

B-2 — CRG ownership locus vs CAP-023 (V-02)

- **Current baseline text (CAP-023):** “For each environment within a subscription and region, reservations are organised into ... one regional CRG plus one per-AZ CRG...” — i.e. CAP-023 as written assumes the CRG lives **in the workload subscription**.
- **What breaks:** The shared prong relocates CRG ownership to a dedicated **provider subscription** that hosts no workload VMs; the workload (consumer) subscription owns no CRG and merely receives a sharing grant. This contradicts the literal “within a subscription” reading of CAP-023 and the assumption baked into three artifacts:
 - `acrme_cr_crg_deployment_layout.md` — shows CRGs co-resident with workloads.
 - `acrme_crg_cr_foundation_provisioning_plan.md` — provisioning steps assume the consumer sub owns the CRG.
 - `acrme_security_and_rbac_guide.md` — CRG owner role is assigned in the workload subscription, not a provider MI.
- Diagrams 7 & 8 (foundation provisioning; sharing model) — provider/consumer split is partial.
- **Important nuance (why it is a real break, not cosmetic):** In the **local** prong the CAP-023 “within a subscription” reading still holds. So CAP-023 must be generalised to say ownership is either the workload subscription (local prong) or a provider subscription (shared prong) — the **structure** (regional + per-AZ, never-mixed) is identical in both. Without this, CAP-023 and CAP-025 (proposed) directly conflict.
- **Fix:** Generalise CAP-023 ownership wording + ratify CAP-025 (provider subscription model, already drafted ADR-007 §15). Gate on POC-006. See T-02.

B-3 — ENV-003 enforcement surface (V-08)

- **Current baseline text (ENV-003):** Non-prod/prod and DR/prod **cannot** share capacity; DR **may** share with non-prod.
- **What “breaks”:** ENV-003’s intent is fully preserved — but the **mechanism** the existing design relies on (don’t co-mingle environments inside one subscription) is **no longer sufficient** on its own. In the shared prong a mis-scoped **sharing grant** (e.g. a Prod provider CRG granted to a subscription that also runs CVAL) creates a cross-environment capacity path that intra-subscription checks would never catch.
- **Specifically what it invalidates:** Any validation/audit design that assumes ENV-003 is satisfied purely by subscription-level environment alignment. A **new mandatory audit** — “no sharing grant crosses the Prod ↔ CVAL/DR boundary” — becomes part of every reconciliation cycle. This is

why **heritage (mixed-environment) subscriptions are excluded from the shared prong** (V-07): granting to them would breach ENV-003 at the grant boundary.

- **Fix:** Ratify CAP-026 (grant lifecycle + boundary audit) and add the coexistence invariant to the reconciliation loop. See T-03.

Everything else does NOT break the existing design — it is either net-new capability (V-03, V-04, V-05, V-06, V-07) or documentation that must be brought forward to match (V-09, V-12, V-13). The **local prong is operationally identical to today's pre-sharing state**, so no currently-running workload is disrupted by adopting the 2-prong model.

5. Remediation Plan — One Task per Variance

Sequenced so that breaking items and their gates land before dependent work. “Gate” = the open POC/decision that must clear before the task can be finalised (design can proceed in parallel).

Task	Addresses	Action	Owner	Gate / Pre-condition	Sequence
T-01	V-01 (B-1)	Apply CAP-013 correction: per-CRG=100 (hard), CRG-is-unit, mandatory workload-domain sharding clause. Update any doc/POC scenario citing “~100/tenant”.	Capacity Eng	none	1
T-02	V-02 (B-2)	Generalise CAP-023 ownership (workload-sub or provider-sub; structure identical). Ratify CAP-025 (provider subscription model).	Platform Arch	POC-006	1
T-03	V-08 (B-3)	Ratify CAP-026 (grant life-cycle + ENV-003 grant-boundary audit). Add grant-boundary check to reconciliation loop.	Security + Platform Eng	DEP-001	1
T-04	V-06	Add INV-R1 (region-	Platform Arch	none	1

Task	Addresses	Action	Owner	Gate / Pre-condition	Sequence
		selection path-agnostic) to ADR-001 principles and ADR-007 §4.			
T-05	V-04	Add MG-scoped sharing-grant clause to CAP-013; design MG-subtree grant propagation.	Platform Eng	POC-003	2
T-06	V-05	Design the shard registry ((domain, region, track)→provider→[CRG] + CRG-type flag local/ shared) and the L2 resolution branch; update UML class model.	Platform Eng	OD-2 (registry store)	2
T-07	V-07	Firm up heritage/modern classification rules (A-01/A-02) and the heritage-exclusion-from-sharing rule; publish remediation backlog format.	Platform Arch	OD-1 (taxonomy)	2

Task	Addresses	Action	Owner	Gate / Pre-condition	Sequence
T-08	V-12	Extend OPS-006/ C-12 with provider-sub + shard-counter tokens (sub-<org>-cld-cr-<track>-<domain>-<region>-<NN>).	Platform Eng	none	2
T-09	V-09	Confirm dedicated-DR-provider-subscription position or revert to shared.	Capacity Eng	POC-006	3
T-10	V-10	Run quota-scope POC; if provider-scoped, keep local-prong fallback (roll-back flag already designed, ADR-007 §14.3).	Capacity Eng	POC-001	3
T-11	V-11	Enterprise Preview-acceptance sign-off or wait for GA before enabling shared prong in prod.	Platform Eng	DEP-001	3
T-12	V-13	Reconcile the 6 design	Platform Arch	T-02, T-03	3

Task	Addresses	Action	Owner	Gate / Pre-condition	Sequence
		docs + 3 diagrams to the dual-path model (provider/consumer split, both paths).			

Critical path: T-01, T-02, T-03, T-04 (the breaking items + the invariant) must complete first — they can proceed in design now; T-02/T-03 finalise only after POC-006/DEP-001. The shared prong cannot be enabled in production until **POC-001 + DEP-001** both clear (T-10, T-11); until then the estate runs the **local prong** with zero disruption.

6. Changes Needed to DESIGN (Documents & Requirements)

6.1 Baseline v2.4 — requirement edits

Req	Change	Type
CAP-013	Replace “~100 per tenant” with “exactly 100 consumer subs per CRG (hard); CRG is the sharing unit; shard by workload domain beyond 100.” Add MG-scope clause.	Correction + addition
CAP-023	Generalise ownership: CRG owned by the workload subscription (local prong) or a provider subscription (shared prong); structure (regional + per-AZ, never-mixed) unchanged.	Correction
CAP-025 (new)	Provider subscription model — infra-only subs own CRGs/CRs; consumers access via grants; shard registry maps scope→provider→CRGs.	New
CAP-026 (new)	Sharing-grant lifecycle — grant/revoke/audit + ENV-003 grant-boundary audit each reconciliation cycle.	New
ENV-003	Add note: in the shared prong, separation is additionally enforced at the sharing-grant boundary (not only intra-subscription).	Clarification
OPS-006 / C-12	Add provider-subscription and shard-counter naming tokens.	Extension
ADR-001 / PLC-006	Add INV-R1 (region selection is path-agnostic).	New invariant

6.2 Design corpus to reconcile (V-13)

- [Reference-Material/reference/acrme_cr_crg_deployment_layout.md](#) (+ docx/pdf)
- [Reference-Material/reference/acrme_crg_cr_foundation_provisioning_plan.md](#) (+ docx/pdf)

- `Reference-Material/guides/acrme_security_and_rbac_guide.md` (provider MI ownership)
- `Design/acrme_uml_class_diagrams_summary.md` (shard registry + CRG-type)
- `Architecture/diagrams/diagram_7_*`, `diagram_8_*` (provider/consumer split; both paths)
- `POC-Test-Scripts/acrme_poc_workbook_v2_4.md` (POC-001/003/006 scenarios reflect per-CRG limit + grant scope)

7. Changes Needed to MEMORY (Standing Instructions / Reference Facts)

Two items warrant a memory update (offered, not auto-applied):

1. **Stale baseline pointer (correctness fix).** The recorded ACRME standing instruction points to `Uploads/...Consolidated Requirements Baseline.md` mirrored at `docs/acrme_requirements_baseline_v2_2.md`. The **current authoritative source is** `Requirements/acrme_requirements_baseline_v2_4.md` (v2.4, 7 Sep 2026). The pointer should be updated so future work re-reads v2.4, not v2.2.
2. **New reference fact (2-prong model canon).** Optionally record: “ACRME CRG deployment = ONE unified structural model (CAP-023) with TWO access paths — shared (provider-subscription-owned CRGs, modern/remediated subs) and local (consumer-owned CRGs, un-remediated heritage). Region selection is path-agnostic (INV-R1). Shared prong is gated on POC-001 + DEP-001.” This prevents the recurring “two models” misframing.

These are **proposed**; I will apply them only on your say-so.

8. Open Decisions Introduced / Reaffirmed

ID	Decision	Why it matters	Owner
OD-1	Confirm interpretation (B) one-model/ two-paths vs (A) two parallel models	(A) would impact region selection and expand V-05/V-07/ V-14	Architecture Board
OD-2	Shard-registry system of record (config repo vs platform DB vs ARG-derived)	Determines T-06 design	Platform Eng
OD-3	Dedicated vs shared DR provider subscription	POC-006; affects T-02/T-09	Capacity Eng
OD-4	Break-glass path when sharing grant fails at scale	Availability of the shared prong	Platform Eng

9. Summary

- The 2-prong model is **one unified structural model with two access paths** — it is “modern-focused” (shared prong is the strategic default) with a **non-disruptive local prong** for heritage. It **does not add complexity to region selection** (guaranteed by INV-R1).
- **13 variances** identified. **Only 3 genuinely break** existing design: **B-1** CAP-013 limit semantics, **B-2** CRG ownership locus vs CAP-023, **B-3** ENV-003 enforcement surface. Each has a specific, bounded fix (T-01/T-02/T-03).
- Everything else is **additive** (new capability) or **doc reconciliation**. **No running workload is disrupted** — the local prong equals today’s behaviour.
- The **shared prong is not production-safe until POC-001 (quota) and DEP-001 (Preview→GA) clear**; design proceeds now, production enablement waits on those gates.
- **Memory**: the recorded baseline pointer is stale (v2.2 → should be v2.4) — recommend updating.